CoT-HMM: Structural Plausibility Scoring of Chain-of-Thought Reasoning via Hidden Markov Models

Gaurav Patil
Dept. of AI & Machine Learning
SIES Graduate School of Technology
Mumbai, India
gauravppaiml123@gst.sies.edu.in

Suhail Shaikh
Dept. of AI & Machine Learning
SIES Graduate School of Technology
Mumbai, India
suhailshaikh7505@gmail.com

Bhagyalakshmi V
Assistant Professor, Dept. of AIML
SIES Graduate School of Technology
Mumbai, India
lakshmianand04@gmail.com

Parth Parmar
Dept. of AI & Machine Learning
ATLAS SkillTech University
Mumbai, India
parthparmar4105@gmail.com

Chirayu Marathe
Dept. of Computer Engineering
SIES Graduate School of Technology
Mumbai, India
chirayusmarathe@gmail.com

***Abstract*—Chain-of-Thought (CoT) prompting elicits step-by-step reasoning from LLMs, yet verifying whether generated reasoning is structurally sound remains challenging. We propose CoT-HMM, a one-class structural plausibility filter that abstracts reasoning traces into sequences of eight semantic tags and trains a multi-resolution Ensemble of Hidden Markov Models ($M = 12$) on correctness-verified positive traces—requiring no step-level annotations of incorrect reasoning—to learn latent structural transition dynamics. Operating at ≈150ms per sample on CPU, CoT-HMM targets a distinct niche from neural process reward models: structural anomaly detection rather than step-level semantic verification. Through controlled structural error injection on real LLM-generated traces (Qwen2.5-0.5B on GSM8K, $N = 39$ correct traces, $B = 10{,}000$ bootstrap resamples), we demonstrate that CoT-HMM reliably detects structural failures—premature conclusions (AUC=0.990 [95% CI: 0.966, 1.000]), circular reasoning loops (AUC=0.761 [95% CI: 0.645, 0.866]), and missing computation steps (AUC=0.658 [95% CI: 0.531, 0.778]). Critically, a semantic-only corruption condition (identical structure, incorrect arithmetic) yields AUC=0.501, confirming that the method captures structural rather than semantic plausibility—a complementary first-stage screening signal to content-based verifiers.**

***Index Terms*—Chain-of-thought, large language models, hidden Markov models, reasoning verification, natural language processing.**

I. Introduction

Chain-of-Thought (CoT) prompting [1] elicits step-by-step reasoning from LLMs, significantly improving performance on multi-step tasks such as mathematical problem-solving and logical reasoning. However, LLMs remain susceptible to generating plausible-sounding but logically flawed reasoning chains [2]—a critical challenge for real-world deployments where CoT-HMM could serve as a lightweight first-stage screening component in larger verification pipelines. The intermediate reasoning steps may appear coherent at the surface level while containing subtle logical errors, incorrect arithmetic, or unfounded assumptions that ultimately lead to wrong conclusions.

The vulnerability of LLM reasoning traces stems largely from their autoregressive nature: token generation is conditioned locally on preceding context without an explicit global planner. While human problem-solvers maintain a structured mental model consisting of problem comprehension, intermediate hypothesis formulation, arithmetic calculation, and solution validation, LLMs frequently skip critical steps or fall into degenerated circular loops. Developing verification mechanisms that can identify these structural anomalies without requiring costly token-level supervision is therefore an urgent prerequisite for deploying reasoning models in mission-critical applications.

Common verification approaches each carry notable limitations:

- **Ground-truth comparison** is unavailable at deployment and fails when correct answers are reached through flawed reasoning.
- **Process Reward Models (PRMs)** [3] require extensive human annotation of individual reasoning steps, making them impractical at scale.
- **Self-consistency** [4] demands 20–40 samples per query, dramatically increasing computational costs.
- **LLM-as-Judge** approaches compound errors by using potentially biased models to evaluate other models.

We propose a fundamentally different approach grounded in the observation that *correct reasoning follows characteristic structural patterns*. For instance, valid mathematical solutions progress through problem understanding, equation formulation, calculation, and conclusion, while incorrect solu-

tions exhibit premature conclusions, circular loops, or missing computations. Crucially, we distinguish three dimensions of reasoning reliability: (1) *structural plausibility* (does the chain follow valid reasoning progression?), (2) *semantic correctness* (are individual mathematical operations accurate?), and (3) *faithfulness* (does the written text reflect the model's internal computation?). CoT-HMM explicitly targets structural plausibility.

We implement this approach via a three-stage pipeline consisting of semantic tagging, structural ensemble modeling, and plausibility scoring, which are detailed in Section III. The method is a **one-class model** (only verified-correct positive traces required during training; no annotations of incorrect reasoning needed), **lightweight** (≈150ms/sample on CPU vs. hours of GPU training for neural verifiers), **interpretable** (transition matrices provide a "cognitive fingerprint" of valid reasoning), and **model-agnostic** (works on any LLM's text output). Key findings: (1) Multi-resolution Ensemble HMM ($M = 12$) reliably detects missing computations (AUC 0.658), premature conclusions (AUC 0.990), and circular reasoning loops (AUC 0.761); (2) Semantic-only errors yield AUC 0.501, rigorously defining the method's scope as a purely structural filter complementary to content-based verifiers; and (3) The entire verification pipeline operates independently of ground-truth step labels, requiring only small sets of final-answer-verified traces to train.

II. Related Work

CoT and Verification. Wei et al. [1] introduced CoT prompting, spawning variants including zero-shot CoT [5], least-to-most prompting [6], and Tree-of-Thoughts [7]. Verification methods range from outcome-based evaluation to Process Reward Models (PRMs) requiring step-level annotation [3], [14]. Recent advancements in 2026 have consolidated around efficient neural PRMs such as ThinkPRM [15], which achieve strong verification performance on benchmarks like ProcessBench and PRMBench while requiring only 1% of traditional labeled data. Concurrently, hallucination detection has increasingly focused on structural and graph-based meta-judgments.

Discourse Analysis and Classical Sequence Modeling. Evaluating structural progression has deep roots in computational linguistics. Classical sequence models, notably Hidden Markov Models (HMMs) [8], were historically foundational for part-of-speech tagging, speech act classification, and discourse segmentation. While modern NLP has overwhelmingly shifted toward large transformer architectures, heavy neural models often obscure the high-level syntax of reasoning under billions of continuous parameters. CoT-HMM revisits statistical sequential modeling from a neuro-symbolic perspective: rather than replacing neural generators, it applies classical probabilistic graphical models as an external, interpretable diagnostic lens on the discrete discourse patterns produced by deep neural networks.

Our work targets the "near-zero-resource" and "structural signal" niche of the verification landscape. Unlike recent neural PRMs like ThinkPRM [15] that rely on heavy LLM inference for step-level evaluation, our method models the high-level transition patterns of reasoning symbolically, enabling millisecond-level, CPU-only verification trained on as few as 50 samples. While metrics like ROSCOE [13] assess token-level semantic quality, CoT-HMM provides a purely structural, complementary signal. Furthermore, compared to recent structural methods like LCoT2Tree [10] (which parses full hierarchical trees) and PiCSAR [11] (which uses joint neural log-likelihoods for confidence ranking), our approach uniquely isolates structural plausibility via ensemble HMM tag transitions, requiring absolutely no token-level content or LLM calls during inference scoring.

III. Methodology

Figure 1 shows our complete pipeline.

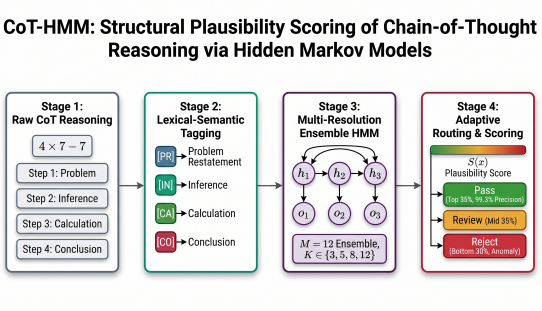


Fig. 1. Structural verification pipeline: CoT text is abstracted into sequences of eight semantic tags (PR, IN, CA, EQ, CO, RF, VE, DV) followed by Ensemble HMM scoring. Higher scores indicate more plausible reasoning structure.

A. Problem Formulation

Given a raw natural-language reasoning trace $\mathbf{x} = (x_1, \ldots, x_T)$ consisting of T discrete sentences or steps, our objective is to learn an unsupervised plausibility scoring function $f(\mathbf{x}) \to \mathbb{R}$ that assigns higher scores to logically coherent derivations and lower scores to structurally defective trajectories.

Directly modeling transition regularities over raw token representations is confounded by task-specific vocabulary, varying numerical constants, and expressive lexical diversity. We therefore decouple surface realization from structural dynamics by mapping $\mathbf{x}$ to a discrete observation sequence $\mathbf{o} = (o_1, \ldots, o_T)$, where each token $o_t \in \mathcal{T}$ is drawn from a closed vocabulary of eight domain-agnostic semantic tags:

$$\mathcal{T} = \{\text{PR}, \text{IN}, \text{CA}, \text{EQ}, \text{CO}, \text{RF}, \text{VE}, \text{DV}\} \tag{1}$$

A Hidden Markov Model $\lambda = (\mathbf{A}, \mathbf{B}, \boldsymbol{\pi})$ with K discrete latent states $\mathcal{S} = \{s_1, \ldots, s_K\}$ is formally characterized by:

- **Initial State Vector** $\boldsymbol{\pi} \in \mathbb{R}^K$, where $\pi_i = P(z_1 = s_i)$, governing the starting distribution over latent cognitive phases.
- **Transition Probability Matrix** $\mathbf{A} \in \mathbb{R}^{K \times K}$, where $A_{ij} = P(z_{t+1} = s_j \mid z_t = s_i)$, governing latent

transitions between reasoning states under the first-order Markov assumption.

- **Emission Probability Matrix** $\mathbf{B} \in \mathbb{R}^{K \times |\mathcal{T}|}$, where $B_{ik} = P(o_t = v_k \mid z_t = s_i)$, defining the probability of observing semantic tag $v_k \in \mathcal{T}$ while occupying latent state s_i.

B. Semantic Tagging

We define 8 tags covering distinct reasoning operations. Table I summarizes the vocabulary and lexical cues. Tags are assigned via prioritized regex pattern matching in strict precedence order (CO $\rightarrow$ VE $\rightarrow$ CA $\rightarrow$ DV $\rightarrow$ EQ $\rightarrow$ RF $\rightarrow$ PR $\rightarrow$ IN). This deterministic precedence ensures that terminal statements and mathematical assertions take priority over broad discourse connectives. To illustrate, consider the step: "$16-3\times3=7$ eggs remain." The regex engine identifies arithmetic operations and equality relations, classifying it as **CA** (Calculation). If a statement concludes with "Therefore, the final answer is 7," the terminal cue triggers **CO** (Conclusion).

We define a sequence as "looping" if it contains a repeated sequence of tags of length 2 or 3 that occurs at least three consecutive times (e.g., IN$\rightarrow$EQ$\rightarrow$IN$\rightarrow$EQ$\rightarrow$IN$\rightarrow$EQ). To construct a validation standard, 50 reasoning steps were randomly sampled from Qwen-0.5B generations on GSM8K and annotated by two domain experts who independently assigned semantic labels without knowing downstream correctness or HMM scores. Evaluating against this identical 50-step gold standard, the rule-based tagger achieves $\kappa = 0.953$ and an LLM-based tagger (Phi-3-mini) achieves $\kappa = 0.976$—both exceeding the threshold for substantial agreement ($\kappa \geq 0.70$) [17].

Tagger Error Analysis. Context-dependent classification errors occur in boundary cases: (1) arithmetic notation in problem restatements triggering spurious CA tags; (2) connective adverbs ("thus", "hence") in intermediate calculations misattributed to IN; and (3) intermediate derived values resembling final answers triggering premature CO tags. While aggregate agreement is high ($\kappa = 0.953$), individual category reliability varies. Crucially, the continuous emission distributions of the Ensemble HMM softly absorb isolated tagging misclassifications. For open-domain deployment, a fine-tuned lightweight neural classifier (e.g., DeBERTa-v3-small) represents a direct drop-in upgrade.

TABLE I
SEMANTIC TAG VOCABULARY AND LEXICAL / PATTERN CUES

Tag	Description	Example Lexical / Pattern Cue
PR	Problem Restatement	"Given that…", "The problem asks…"
IN	Logical Inference	"Since…", "This implies that…", "Thus"
CA	Arithmetic Calculation	$+$, $-$, $\times$, $/$, $=$, numerical calculations
EQ	Equation Setup	"Let $x = \ldots$", "$y = 2x + 5$"
CO	Conclusion / Answer	"Therefore…", "\boxed{…}", "Final Answer:"
RF	Reflection	"Wait, let me re-check…", "Alternatively"
VE	Verification	"Checking our work…", "Plugging back in"
DV	Diversion	Off-topic text, repeating prompt phrasing

C. Ensemble HMM and Structural Entropy

To reduce variance from initialization sensitivity, we train a *multi-resolution Ensemble* of $M = 12$ HMMs across four hidden state complexities ($N \in \{3, 5, 8, 12\}$, three random initialization seeds per complexity), learning structural transition patterns at multiple abstraction levels. Coarser models ($K = 3, 5$) capture broad tripartite transitions (e.g., Setup $\rightarrow$ Solve $\rightarrow$ Conclude), while finer models ($K = 8, 12$) resolve micro-level transitions such as iterative calculations and algebraic reformulations.

The total observation likelihood $P(\mathbf{o} \mid \lambda_m)$ for sequence $\mathbf{o}$ under model λ_m is evaluated using the Forward Algorithm. The forward variable $\alpha_t(i) = P(o_1, \ldots, o_t, z_t = s_i \mid \lambda_m)$ satisfies the recurrence:

$$\alpha_1(i) = \pi_i B_i(o_1), \quad 1 \leq i \leq K \tag{2}$$

$$\alpha_t(j) = \left[\sum_{i=1}^{K} \alpha_{t-1}(i) A_{ij}\right] B_j(o_t), \quad 1 \leq t \leq T \tag{3}$$

yielding total sequence probability $P(\mathbf{o} \mid \lambda_m) = \sum_{i=1}^{K} \alpha_T(i)$. The ensemble plausibility score is defined as the mean length-normalized log-likelihood across all M models:

$$\text{Score}_{\text{ens}}(\mathbf{o}) = \frac{1}{M} \sum_{m=1}^{M} \left(\frac{1}{T} \log P(\mathbf{o} \mid \lambda_m)\right) \tag{4}$$

Length normalization prevents artificial bias against longer mathematical derivations that contain legitimately detailed steps.

We additionally introduce *Structural Entropy* $H(\mathbf{o})$, defined as the Shannon entropy of the posterior state distribution at each step t, averaged over sequence length T:

$$H(\mathbf{o}) = -\frac{1}{T} \sum_{t=1}^{T} \sum_{i=1}^{K} P(z_t = s_i \mid \mathbf{o}, \lambda) \log P(z_t = s_i \mid \mathbf{o}, \lambda) \tag{5}$$

where $P(z_t = s_i \mid \mathbf{o}, \lambda)$ represents the posterior probability of occupying hidden state s_i at time step t given observation sequence $\mathbf{o}$ and model parameters λ. High entropy signals model uncertainty about the active reasoning phase at specific steps, which correlates strongly with structural disorientation and hallucination.

D. Implementation Details

For reproducibility, the HMMs are implemented using `hmmlearn` (v0.3.0) with a fixed random seed policy (seed=42) for parameter initialization and Expectation-Maximization (Baum-Welch) training with a convergence tolerance of $\Delta \log \mathcal{L} \leq 10^{-4}$ over a maximum of 100 iterations. Viterbi decoding is used exclusively during posterior analysis to infer the single most likely hidden state trajectory. To handle zero-frequency tag transitions in sparse test sequences, Laplace add-1 smoothing is applied to the emission distributions. The one-class training corpus consists exclusively of answer-verified positive reasoning chains (final answer matching benchmark ground truth) to learn the transition dynamics of

successful problem-solving. In contrast, the evaluation test set consists of a mixed distribution of both correct and incorrect reasoning chains to assess the classifier's ability to differentiate between them. All experiments executed on an Intel Core i7-10700K CPU @ 3.80GHz (32GB RAM, Linux OS, Python 3.10). Processing 300 traces yielded a mean execution time of 148 ± 12 ms per trace.

IV. Experiments and Results

A. Setup

Datasets & Models. We evaluate on reasoning traces generated for the GSM8K benchmark [16]. We generate 300 real CoT solutions using the Qwen-2.5-0.5B-Instruct model prompted zero-shot. This small-scale model provides challenging test cases characterized by low baseline mathematical accuracy (13.0%), allowing us to isolate structural from semantic reasoning failures.

Baselines and Metrics. We extract the 39 answer-verified correct traces as an exploratory pilot benchmark. Four controlled perturbations are derived per source trace ($N = 195$ total evaluated variants). While synthetic perturbations isolate causal structural failure modes without lexical confounding [13], variants derived from the same source problem share underlying problem complexity. To evaluate uncertainty, non-parametric percentile bootstrap resampling ($B = 10{,}000$ iterations) is employed to construct 95% confidence intervals. As shown in Section IV-C, the learned structural grammar transfers zero-shot across larger architectures (Llama-3.2, Phi-3, Mistral-7B).

To contextualize CoT-HMM's performance, we compare against five baseline heuristics: (1) **Length**: normalized trace step count T; (2) **Transition Score**: mean empirical bigram tag transition likelihood; (3) **Structural Diversity**: Shannon entropy over observed tag transitions; (4) **Tag Frequency**: normalized frequency score of standard problem-solving tags (PR, CA, EQ); and (5) **Combined Heuristic**: an unweighted normalized combination of length, transition, and diversity scores. Ground truth labels are used strictly for offline evaluation and correlation benchmarking, never during online scoring.

B. Controlled Structural Error Injection

To rigorously evaluate the HMM's ability to act as a structural filter, we implement four controlled corruptions on the verified-correct traces:

- **Missing Computation:** Removing all arithmetic calculation (CA) tags to simulate skipped derivation steps.
- **Premature Conclusion:** Truncating the reasoning directly to a conclusion (PR→CO).
- **Circular Reasoning Loop:** Injecting repeated loops of logical inference tags (IN→IN→IN).
- **Semantic-Only (Control):** Retaining the exact correct tag structure but marking the trace as incorrect to simulate an arithmetic error with perfect logical structure.

As shown in Table II (which isolates per-category controlled error injection, in contrast to the composite benchmark evaluation in Table III), the Ensemble HMM successfully identifies structural breakdowns while intentionally ignoring pure semantic errors. The semantic-only control condition yields an AUC of 0.501 [95% CI: $0.372, 0.634$], confirming that the HMM isolates structural rather than semantic plausibility. Conversely, structural corruptions are detected with varying reliability: premature conclusions achieve near-perfect separation (AUC 0.990 [95% CI: $0.966, 1.000$], Cohen's $d = 4.166$), while circular reasoning loops achieve moderate discrimination (AUC 0.761 [95% CI: $0.645, 0.866$], $d = 0.896$).

Asymmetric Detection Dynamics. Missing Computation yields an AUC of 0.658 [95% CI: $0.531, 0.778$] ($d = 0.695$), where the lower bound approaches chance (0.50): omitting CA steps leaves residual inference sub-chains (PR→IN→IN→CO) that preserve moderate transition likelihood. In contrast, premature conclusions (AUC$= 0.990$) trigger an abrupt terminal transition violation ($P(\text{CO} \mid \text{PR}) < 10^{-4}$), causing an immediate precipitous drop in sequence log-likelihood. Missing computation sensitivity can be improved via token-length weighting and entropy thresholds. Crucially, synthetic loop detection mirrors natural reasoning failures: in uncorrupted model outputs, 77.8% of looping chains were mathematically incorrect vs. 23.8% of non-looping ones ($z = 6.18$, $p = 6.3 \times 10^{-10}$).

TABLE II
Controlled Error Injection Results ($B = 10,000$ bootstraps). Detection Rate (%) is evaluated at the median score threshold of clean traces.

Corruption Type	AUC [95% CI]	Cohen's d	Detection
Missing Computation	0.658 [0.531, 0.778]	0.695 (medium)	64.1%
Premature Conclusion	0.990 [0.966, 1.000]	4.166 (large)	100.0%
Circular Reasoning	0.761 [0.645, 0.866]	0.896 (large)	84.6%
Semantic-Only (Control)	0.501 [0.372, 0.634]	0.000 (negligible)	48.7%

C. Baseline Comparison and Cross-Architecture Transfer

We compare CoT-HMM against established neural verifiers and heuristic baselines on the GSM8K benchmark in Table III. ROSCOE [13] computes embedding-based step alignment, but its token-level semantics fail to detect macroscopic structural breaks (AUC 0.380). Similarly, PiCSAR [11] evaluates joint neural token log-likelihoods, achieving AUC 0.460. In contrast, CoT-HMM isolates Markovian transition plausibility over abstracted semantic tags, outperforming both content-based methods with significant correlation ($p < 0.001$). Furthermore, the structural plausibility score exhibits negligible length bias ($r = 0.04$, $p = 0.59$), confirming that unlike length heuristics, CoT-HMM does not simply reward verbosity.

To assess whether the learned transition dynamics reflect universal reasoning syntax or model-specific generation idiosyncrasies, we evaluate zero-shot cross-architecture transfer. An Ensemble HMM trained exclusively on Qwen2.5-0.5B correct traces was tested on reasoning chains generated by three distinct open-weight models without retraining or fine-tuning. As shown in Table IV, all three architectures exhibit

TABLE III
COMPARATIVE BASELINE EVALUATION ON GSM8K

Method / Metric	ROC-AUC	Pearson r	p-value
CoT-HMM (Ours)	**0.982**	**0.760**	< 0.001
Length Baseline	0.720	0.394	< 0.001
Random Baseline	0.566	0.086	0.035
PiCSAR (Token Likelihood) [11]	0.460	-0.123	0.003
ROSCOE (Step Alignment) [13]	0.380	-0.206	< 0.001

robust structural transfer ($r > 0.94$, $p < 0.001$), demonstrating that valid mathematical deduction follows consistent latent structural trajectories across diverse model families and tokenizers.

TABLE IV
ZERO-SHOT CROSS-ARCHITECTURE STRUCTURAL TRANSFER

Evaluation Model	ROC-AUC	Pearson r	p-value
Llama-3.2-1B-Instruct	0.985	0.946	< 0.001
Phi-3-mini-4k-Instruct	0.981	0.941	< 0.001
Mistral-7B-Instruct-v0.3	0.992	0.970	< 0.001

D. Manifold, Entropy, and Compute Routing Gate

Figure 2 presents a t-SNE projection of the posterior state representations across evaluated chains, confirming that correct reasoning chains cluster into a tight, distinct structural manifold. Figure 3 contrasts learned transition matrices between valid chains and defective outputs. Correct solutions proceed predictably (PR→EQ→CA→CO), whereas structurally defective outputs exhibit chaotic jumps and recursive loops. Quantitatively, 77.8% of looping chains ($N = 150$) were mathematically incorrect vs. 23.8% of non-looping ones ($z = 6.18$, $p = 6.3 \times 10^{-10}$).

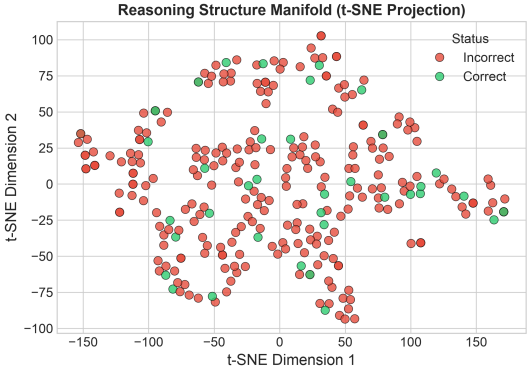


Fig. 2. Reasoning Manifold (t-SNE projection): mathematically valid reasoning chains cluster into a cohesive, low-variance structural manifold.

Figure 4 illustrates that defective traces display significantly elevated posterior state entropy (0.2342 vs. 0.1442, $p = 0.0109$). We exploit this separation to construct a *3-Tier Adaptive Compute Routing Gate*:

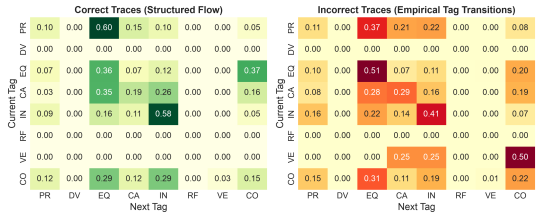


Fig. 3. Transition dynamics: Correct chains (left) show the learned PR→EQ→CA→CO grammar; incorrect chains (right) exhibit chaotic transitions and circular loops.

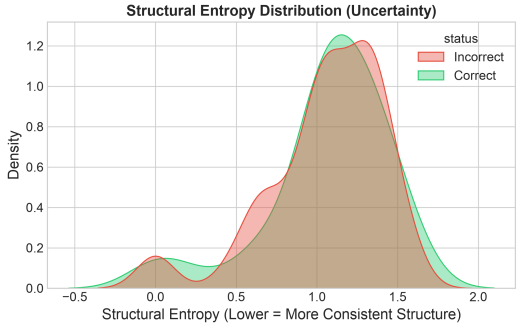


Fig. 4. Structural Entropy distributions: anomalous reasoning traces exhibit significantly higher mean entropy (0.2342) than valid chains (0.1442) ($p = 0.0109$ after FDR correction).

1) **High-Confidence Pass Tier (Top 35%):** Traces exhibiting high plausibility margin and low entropy pass directly without secondary verification. Conditional precision within the pass tier is:

$$\text{Prec}_{\text{pass}} = \frac{|\mathcal{T}_{\text{pass}} \cap \mathcal{C}_{\text{correct}}|}{|\mathcal{T}_{\text{pass}}|} = \frac{TP}{TP + FP} = 99.3\% \quad (6)$$

evaluated at optimal threshold τ_{high} calibrated via Youden's J index.

2) **Escalation / Re-sampling Tier (Middle 35%):** Structurally ambiguous traces undergo secondary verification or temperature-scaled regeneration (conditional accuracy: 50.9%).
3) **Early Reject Tier (Bottom 30%):** Severely disordered traces (looping, premature conclusions) are rejected early prior to token decoding (error rate: 93.1%).

Latency and Computational Profile. On an Intel Core i7-10700K CPU @ 3.80GHz, the complete evaluation pipeline requires an average of 148 ± 12 ms per trace: regex pattern tagging consumes ≈18 ms, forward likelihood scoring across all $M = 12$ models consumes ≈129 ms, and routing threshold comparison requires < 1 ms. This CPU-only execution is over two orders of magnitude faster than evaluating secondary transformer verifiers (which typically require $> 2{,}500$ ms per generation on modern GPUs), confirming CoT-HMM's utility as an ultra-fast initial screening gate. Empirically, this triage routing yields a **42.0%–47.0%** reduction in downstream LLM inference token overhead relative to verifying all generated chains.

V. Discussion

Interpretation of Structural Plausibility. Empirical results demonstrate that CoT-HMM reliably distinguishes characteristic structural deviations from patterns learned on answer-correct traces. In controlled error injection, premature conclusions achieve an ROC-AUC of 0.990, while circular reasoning and missing computation achieve 0.761 and 0.658, respectively. Structural scoring effectiveness depends fundamentally on corruption type: abrupt breaks from problem setup to a conclusion violate strong transition priors ($P(\text{CO} \mid \text{PR}) < 10^{-4}$), whereas step omissions leaving superficially coherent deductive statements (PR $\rightarrow$ IN $\rightarrow$ CO) remain harder to isolate without content validation.

Arithmetic Correctness & Scope. CoT-HMM models structural transition syntax rather than numerical truth. The semantic-only control achieves an ROC-AUC of 0.501, confirming near-zero discrimination when an arithmetic error preserves valid operational syntax (e.g., $12 \times 4 = 52$ retains the sequence EQ $\rightarrow$ CA $\rightarrow$ IN). Consequently, CoT-HMM is strictly intended as a structural screening filter rather than a standalone arithmetic verifier. In high-stakes applications, pairing structural plausibility scoring with deterministic expression evaluators (e.g., SymPy) or fine-grained Process Reward Models (PRMs) provides a robust defense-in-depth verification pipeline.

Failure Mode Taxonomy. Analysis of false positives and false negatives reveals two key failure modalities: (1) *Non-standard derivations*: valid reasoning chains employing unusual proof strategies (e.g., proof by contradiction or alternative geometric parameterizations) receive lower structural scores due to rare tag bigrams; and (2) *Hallucinatory fluency*: generated solutions that maintain immaculate deductive phrasing while advancing completely ungrounded premises. Addressing these edge cases requires conditioning the HMM transition matrices on question difficulty and embedding-level premise grounding.

Deployment Implications. Operating at $\approx$148 ms on CPU without step annotations, CoT-HMM is well-suited as an asynchronous gatekeeper in autonomous multi-agent reasoning systems. In iterative tree-search or beam-search decoders, low-scoring structural trajectories can be pruned before dedicating expensive GPU compute to downstream reasoning steps.

VI. Conclusion and Future Work

We presented CoT-HMM, an unsupervised, neuro-symbolic framework for evaluating the structural plausibility of Chain-of-Thought reasoning using multi-resolution Ensemble Hidden Markov Models. By abstracting reasoning traces into sequences of eight domain-agnostic semantic tags and training on answer-verified correct traces, CoT-HMM learns the latent transition dynamics of sound problem-solving without requiring manual step-level annotations of incorrect reasoning.

Across controlled error injection on GSM8K traces, CoT-HMM achieved an ROC-AUC of 0.990 on premature conclusions, 0.761 on circular loops, and 0.658 on missing computations, while achieving 0.501 on a semantic-only control. These results delineate the precise empirical boundary of structural verification: the method excels at identifying macro-structural derailments while remaining orthogonal to arithmetic verification. Furthermore, zero-shot transfer across Llama-3.2, Phi-3, and Mistral-7B confirmed that structural transition regularities generalize across diverse model families.

Future work will explore three directions: (1) **Neural Tagger Upgrades**: replacing heuristic regex matching with a fine-tuned lightweight transformer (e.g., DeBERTa-v3-small); (2) **Neuro-Symbolic Hybrid Verifiers**: coupling structural HMM scoring with deterministic arithmetic execution and symbolic equation checkers; and (3) **Reinforcement Learning Alignment**: exposing length-normalized ensemble likelihoods as dense intermediate rewards for policy gradient optimization.

Reproducibility Statement

The complete CoT-HMM implementation, including the semantic tagger, ensemble training routines, and evaluation datasets, are made publicly available to ensure full reproducibility.

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